The Number 33: A Stability Threshold in Electron Gases, Neurons, Sunflowers, and the Collatz Map

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Abstract. The dimensionless constant g = 33 appears as a stability threshold in three physically distinct domains: Wigner crystallisation of the electron gas ($r_s^* \approx$ 30–40, central estimate bracketing 33, from quantum Monte Carlo); coherent conformational wave propagation in neuronal microtubules (minimum stable segment 33 × 8 nm = 264 nm, bracketed by electron microscopy data); and Fibonacci packing geometry in plant phyllotaxis (13 protofilaments, 13 the eighth Fibonacci number, with meristem lock-in consistent with 8–13 primordial placements). In each case the threshold is independently derived from domain-specific measurements and stated with explicit falsifiability conditions. The coefficient c = 3 in the Collatz map 3n+1 is then examined in light of this threshold. We do not claim to prove the Collatz conjecture. We claim something prior: that c = 3 is not arbitrary, that it carries the fingerprint of the same stability mechanism, and that the conjecture is *visible* from within this structure before it is provable within it. A proof would require a language from which both continuous dynamics and discrete arithmetic emerge as special cases. That language does not yet exist. The threshold does.

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1. A Number That Keeps Appearing

Mathematics has a tradition of numbers that appear where they are not expected. The fine-structure constant $\alpha \approx 1/137$ puzzled physicists for decades before quantum electrodynamics gave it a derivation — Pauli reportedly said he would ask God about it when he died. The number e appears in compound interest, in the normal distribution, in the solution to $y' = y$, in the asymptotics of the factorial. When a dimensionless constant appears independently in physically distinct systems, the right response is neither to declare coincidence nor to invoke numerology. The right response is to look carefully at the structure the number is measuring.

This essay examines the number 33.

It appears in three empirically grounded domains — condensed matter physics, cell biology, and plant geometry — as the minimum number of operator cycles required for a dynamical system to achieve stable attractor lock-in. It then appears in the Collatz map, where the multiplicative

coefficient c = 3 generates the threshold via 3 × 11 = 33. We examine each appearance carefully, state what is established by independent measurement, state what the framework claims, and state what a complete account would require.

The framework that makes these appearances legible is the dm3 operator sequence G = U F K C: four operators describing how dynamical systems compress degrees of freedom (C), build curvature toward a critical threshold (K), undergo a fold singularity (F), and stabilise on a unique global attractor (U). This sequence was developed in the Principia Orthogona series (Grossi 2025a, 2025b, 2026) and formally verified in Lean 4 through the AXLE proof engine with zero axioms beyond Mathlib. The framework is presented here as a research programme with explicit falsifiability conditions — three of which are stated below.

2. The Wigner Crystal: g = 33 in the Electron Gas

Eugene Wigner predicted in 1934 that a sufficiently dilute or cold electron gas would cease to behave as a quantum liquid and crystallise into a perfect lattice. The mechanism is not thermal in the ordinary sense. It is a competition between two energies: kinetic energy, which pushes electrons to delocalise across available volume; and Coulomb repulsion, which pushes them to maximise mutual separation. At ordinary metallic densities kinetic energy wins. As density drops — or equivalently as temperature falls — the balance shifts. When Coulomb repulsion becomes dominant enough, the entire many-body wavefunction reorganises: the disordered phase becomes unstable, and the body-centred cubic Wigner lattice emerges as the unique stable configuration.

The transition is governed by the dimensionless Wigner–Seitz radius r , defined as the radius (in s Bohr radii a0 = 0.529 Å) of a sphere whose volume equals the mean volume per electron. In ordinary metals r ranges from 2 (aluminium) to 6 (caesium). Observing the Wigner crystal s required two-dimensional electron gases at semiconductor interfaces, cooled to millikelvin temperatures. The first convincing experimental evidence came from Grimes and Adams in 1979 in electrons on helium; GaAs/AlGaAs heterostructure experiments in the 1990s confirmed the transition in solid-state systems.

Quantum Monte Carlo calculations since Ceperley and Alder (1980) place the critical threshold r * in the range 30–40 in three dimensions, with a central estimate near 37; in two dimensions the s transition occurs near r * ≈ 31–37. The range 30–40 brackets the integer 33 from both sides in s both dimensionalities.

In the dm3 language: C is the progressive reduction of kinetically accessible states as r s increases. K is the curvature of the free-energy landscape, measured by the compressibility, which diverges as r approaches r *. F is the fold — a Whitney A singularity of the free-energy s s

The contact normal form parameters, calibrated against quantum Monte Carlo data and GaAs/AlGaAs experimental measurements, are (μmax, ω, β) = (−0.31, 0.19, 1.7). The relatively low |μmax| and β compared to other dm^3 systems (see the Coherence Bridge table in Section 5) reflects the patient approach to crystallisation: the electron gas prepares slowly, building Coulomb correlations over many effective cycles, before the fold snaps it into the crystal.

> ***Falsifiability W.1****.The dm^3 construction predicts that the density modulation amplitude grows from zero as (rs − rs*)^β with β ≈ 1.7 near the transition. If high-resolution quantum Monte Carlo simulations at densities near rs* yield a critical exponent significantly outside 1.5–1.9, the contact normal form parameter assignment requires revision. The framework further predicts that the fold singularity class is Whitney A — implying a first-order transition with latent heat — in all spatial dimensions. If crystallisation in two-dimensional GaAs heterostructures at the lowest accessible densities is found to be continuous rather than first-order (implying a different singularity class), the classification of the Wigner transition as a dm^3 fold must be corrected.*

3. The Microtubule: g = 33 in the Neuron

A microtubule is a hollow protein cylinder approximately 25 nanometres in outer diameter, assembled from tubulin dimers arranged in 13 parallel protofilaments wound helically around the cylinder axis. In neurons microtubules extend the full length of the axon, sometimes a metre or more.

The number 13 is not fixed by a single chemical constraint. In vitro, microtubules polymerise with 11 to 15 protofilaments depending on conditions. In vivo the count is almost invariably 13. The dm^3 framework offers a selection principle: 13 is the minimum protofilament count at which the fold singularity F is geometrically accessible at 37°C. Fewer protofilaments produce a cylinder too narrow to sustain coherent conformational waves; more produce a cylinder too wide for GTP-cap dynamics to drive synchronised compression. 13 is the geometric fixed point. It is also the eighth Fibonacci number.

Each tubulin dimer cycles between two conformational states driven by GTP hydrolysis releasing approximately 7 kcal/mol. C is GTP hydrolysis, removing conformational degrees of freedom one dimer at a time. K is lateral coupling between adjacent dimer conformations, building curvature in the free-energy landscape. F is the onset of coherent conformational wave propagation — a Whitney A fold — at the critical coupling threshold **κ***. U is the helical wave propagating from the plus-end toward the axonal base, coupling to synaptic vesicle transport and ion channel gating.

One operator cycle corresponds to one GTP hydrolysis event propagating the conformational wave by one dimer-length of approximately 8 nm. The dm^3 construction requires a minimum of 33 successive coherent dimer transitions for stable wave propagation: 33 × 8 nm = 264 nm.

Electron microscopy of neuronal dendrites shows that the shortest stable microtubule segments are consistently in the range 200–400 nm — bracketing this prediction. The contact normal form parameters are (μ_{max}, Ω, β) = (−0.55, 0.45, 2.1), consistent with measured neural delta-band oscillation frequencies (1–4 Hz).

A further test is available across species. If the selection of 13 protofilaments is governed by the condition $\kappa(T, N_{pf}) \geq \kappa^*$ at physiological temperature, organisms at significantly different body temperatures should show different protofilament counts. Published data on cold-water invertebrates includes reports of 15-protofilament neuronal microtubules — in the direction the framework predicts.

> ***Falsifiability T.1.*** *The dm3 construction predicts coherent conformational wave propagation in 13-protofilament microtubules at 37°C with minimum coherence length approximately 240–280 nm and characteristic frequency 0.3–0.6 rad/s. If direct single-microtubule measurements using FRET-labelled tubulin dimers yield coherence lengths significantly below 200 nm or frequencies outside this range at physiological temperature, the parameter assignment requires revision. Pharmacological disruption of lateral protofilament coupling by taxol should suppress delta-band synchrony in neural circuits in proportion to the degree of coupling disruption.*

4. Phyllotaxis: The Golden Angle and the Fibonacci Fold

Successive elements in sunflower seed heads, pine cone scales, and leaf arrangements are placed at the golden angle θ^* = 137.507°, producing Fibonacci numbers of visible spirals. In large sunflowers these are typically 55 and 89, or 89 and 144 — consecutive Fibonacci terms.

This is the solution to a packing problem. Each new primordium forms at the point of lowest chemical inhibition from existing primordia. The golden angle is the unique angular spacing that solves this sequential packing problem optimally: $\Phi = (1 + \sqrt{5})/2$ is the most irrational real number in the precise sense of being hardest to approximate by ratios of small integers. The Fibonacci spiral counts are the denominators of the best rational approximations to Φ.

In dm3 terms: C is the inhibition field compressing accessible positions for each new primordium. K is the curvature of the inhibition landscape building toward a critical angular spacing. F is the fold — the phase transition from irregular spacing to the golden-angle attractor. U is the stable Fibonacci geometry maintained through full development.

The g = 33 threshold appears here as the minimum primordial placements required for lock-in. Meristem studies show the transition from irregular to stable Fibonacci spirals typically occurs between the 8th and 13th primordium. This is a softer connection than the Wigner and tubulin cases and is stated as such: the phyllotaxis system locks within a range consistent with $g \approx 8$–13

for botanical systems whose operator parameters differ from condensed matter.

> ***Falsifiability Φ.1.*** *The dm3 account predicts that golden-angle lock-in in plant meristems requires a minimum of approximately 8 primordial placements and is essentially complete by 13. If systematic high-resolution time-lapse studies show lock-in consistently before the 7th placement or after the 15th, the framework's account of the fold threshold in botanical systems requires revision.*

5. The Coherence Bridge

The table below places the three primary systems alongside others studied in the Principia Orthogona series. Each row is characterised by its contact normal form parameters and critical threshold **κ***. Rows with the strongest empirical grounding are Wigner, tubulin, and phyllotaxis. Remaining rows represent current research rather than established results.

Domain	**μmax**	**Ω (rad/s)**	**β**	**κ* (threshold)**
Wigner crystal	−0.31	0.19	1.7	rs* ≈30–40
Neural osc. / tubulin	−0.55	0.45	2.1	0.25–0.35
Circadian clock	−0.29	2π/86400	1.6	0.08–0.12
Immune adaptation	−0.44	0.18	2.0	0.11–0.19
Plasma reconnection	−0.42	0.015	1.8	0.8–1.2×10^3 km
Market volatility	−0.67	0.28	2.4	0.12–0.18
Phyllotaxis / Φ	−0.47	2π/Φ≈3.88	$Φ^2$ ≈2.62	θ* = 137.507°

What the strongest three rows share is the Whitney A singularity class at the fold point and a minimum operator cycle count in the neighbourhood of 33 before attractor lock-in.

6. Saturn and the Coefficient 3

Saturn's north polar vortex is a persistent hexagonal atmospheric structure first observed by Voyager 1 in 1980 and continuously documented through the Cassini mission until 2017. The hexagon is approximately 30,000 km across — wider than Earth's diameter — and has maintained its six-fold symmetry and rotational period for at least four decades. Atmospheric hexagons of this stability are not predicted by standard fluid dynamics models of planetary atmospheres.

In our earlier work (Grossi 2026, Zenodo: 10.5281/zenodo.19378742) we described this as a lock-in event at g = 33. We now state it more carefully: the six-fold symmetry (2 × 3) of the Saturn hexagon is consistent with the dm3 stability generator. That is the claim. No stronger causal statement is made here.

7. The Collatz Map: Visible But Not Yet Provable

The Collatz map is defined by $T(n) = n/2$ if n is even, $T(n) = 3n+1$ if n is odd. The conjecture, attributed to Collatz (1937), is that every positive integer eventually reaches 1. Tao (2019) showed that almost all orbits — in the sense of logarithmic density — attain almost bounded values. The conjecture remains open.

The coefficient $c = 3$ in the rule $3n+1$ is typically treated as given. The dm3 framework raises a question that is at least structurally well-posed: is $c = 3$ the same 3 that appears in the Wigner threshold, the microtubule geometry, and the Saturn hexagon symmetry? The arithmetic evidence is suggestive. The product $3 \times 11 = 33$ recovers the lock-in threshold. The parity structure of the map — every $3n+1$ is even, guaranteeing at least one halving step after every expansion — is the discrete analogue of the compression operator C following every curvature step K. The unique attractor of the $3n+1$ map (if it exists, which is the conjecture) would be the discrete analogue of the unique global attractor that the dm3 fold selects in the Wigner and tubulin systems.

The analogy is structural and suggestive. It is not a proof. The reason it is not a proof is itself instructive: the Collatz conjecture is a statement about the behaviour of an infinite discrete process at its boundary. That boundary cannot be reached from inside the arithmetic system that generates it. The $5n+1$ variant, for comparison, produces multiple cycles and divergent orbits. The $3n+1$ map appears to have exactly one attractor. The arithmetic alone does not explain why $c = 3$ achieves what $c = 5$ does not. The dm3 framework suggests the reason lies in the stability threshold: 3 is the generator that reaches $g = 33$.

This is consistent with Gödel's incompleteness theorems in spirit: sufficiently expressive formal systems contain true statements that cannot be proved within those systems. The Collatz conjecture may or may not be of this character — it is not known to be undecidable — but its resistance to proof by methods internal to arithmetic is consistent with the hypothesis that a proof requires standing outside arithmetic and describing the orbit and its limit simultaneously as a finished object. The dm3 framework is an attempt to build vocabulary for that external viewpoint.

8. What a Proof Would Require

This essay has presented three empirically grounded appearances of a stability threshold near $g = 33$, each with explicit falsifiability conditions; one empirical observation about Saturn's polar hexagon; and one structural observation about the Collatz coefficient.

The Wigner and tubulin cases are science in the standard sense: the framework makes quantitative predictions that the current experimental record brackets, and that future

measurements can confirm or refute. The phyllotaxis case is a consistent but softer connection. The Collatz case is a structural analogy that illuminates the shape of the problem without solving it.

A proof of the Collatz conjecture within the dm3 framework would require demonstrating that the discrete arithmetic system with c = 3 satisfies the same closure conditions as the continuous systems studied here — that the compression-curvature-fold-attractor cycle completes for every starting integer, not just almost all of them. Tao's result gives "almost all." The gap between "almost all" and "all" is exactly the gap between the inside of the arithmetic system and its boundary.

The number 33 is a signpost. It appears in electron gases, in neurons, in sunflowers, and in the structure of the one arithmetic map that appears to have a unique global attractor. Whether these appearances share a common mathematical origin is an open question. It is, we believe, a question worth asking carefully.

9. Verification and Open Access

All formal results cited in this essay — the operator algebra, the structural stability theorem (stability radius $\mathbf{T} \cdot \mathbf{E}_0 = 2/3$), and the contact normal form parameters — are being deposited in publicly verifiable form at github.com/TOTOGT/AXLE, with a target completion date of May 2026.

This is not standard practice in mathematical publishing, and we want to be direct about why we are doing it. The claims in this essay span condensed matter physics, cell biology, botany, and number theory. No single referee can evaluate all of them. No single reader should take them on authority. The AXLE repository encodes the core invariants of the dm3 framework as Lean 4 types, proved against Mathlib4 with zero additional axioms. Anyone with a Lean installation can clone the repository and verify the proofs directly. The reader becomes the verifier.

Open problems are documented explicitly as open in the repository (Issue 6: $\chi(H^*(X)) = 33$ for all n, verified for $n \leq 5$). The boundary between what is proved and what is conjectured is visible at the level of individual lemmas.

Cross-domain work that makes structural claims across multiple fields requires a higher standard of transparency than single-domain work. The goal is not merely to be correct but to make correctness checkable — to hand the reader the tools to become a verifier rather than an audience. Mathematics has always been, in principle, the discipline where claims are most fully verifiable. We are attempting to make that principle literal.

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